LoRA: Low Rank Adaptation of Large Language Models

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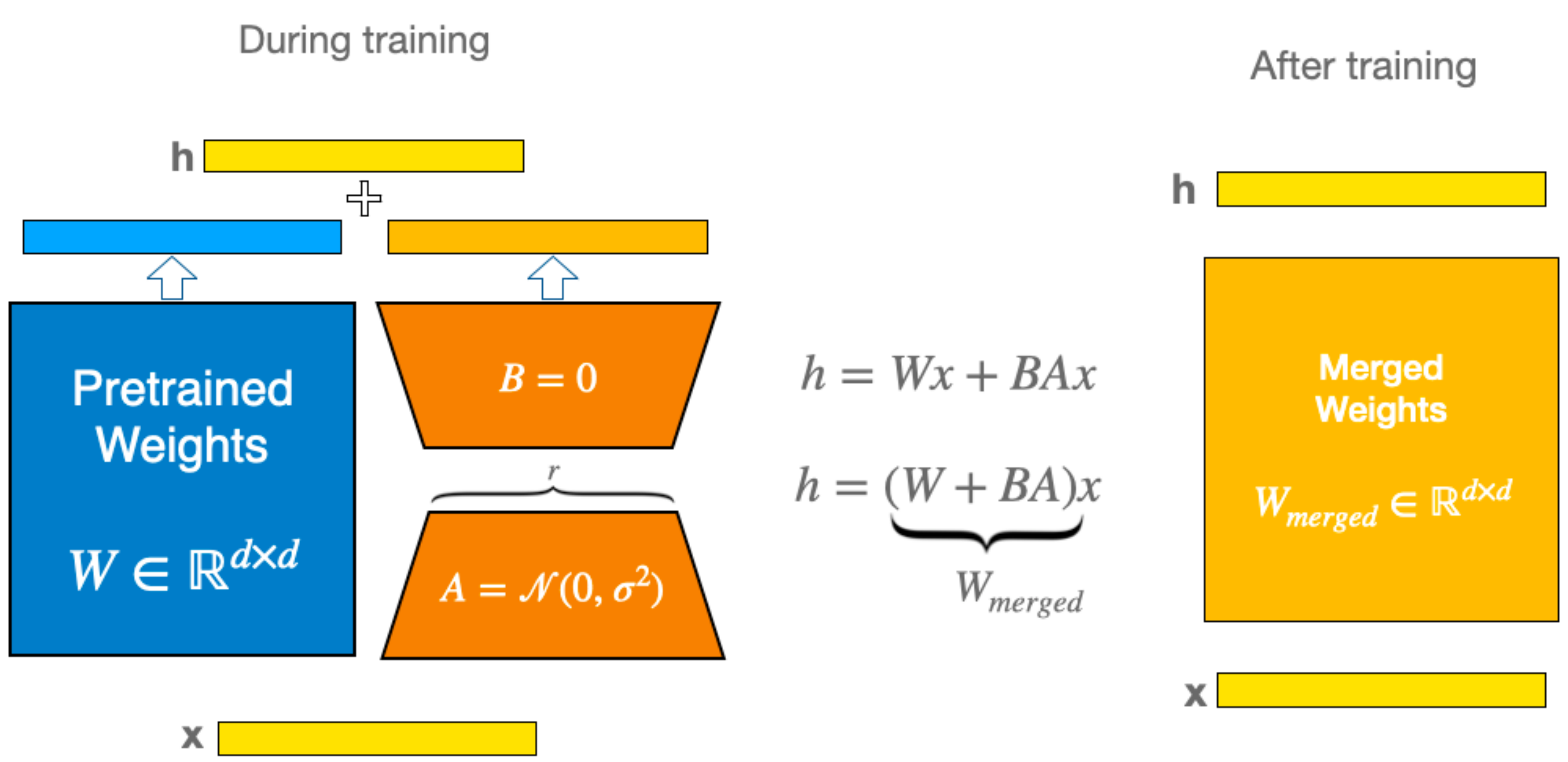
Introduction

- **Problem**
  - Full fine-tuning updates all model parameters.
  - This is expensive in memory, storage, and compute.
  - It makes multi-task adaptation/deployment less practical.
- **Goal**
  - Reproduce LoRA (Hu et al., ICLR 2022).
  - Test whether low-rank adaptation can match/exceed full fine-tuning.
  - Verify this while training far fewer parameters.
- **What We Evaluated**
  - GLUE classification (BERT-base):
    - SST-2
    - MRPC
  - Language modeling (GPT-2 small):
    - WikiText-2 (perplexity)

How can we exploit low-rank structure?

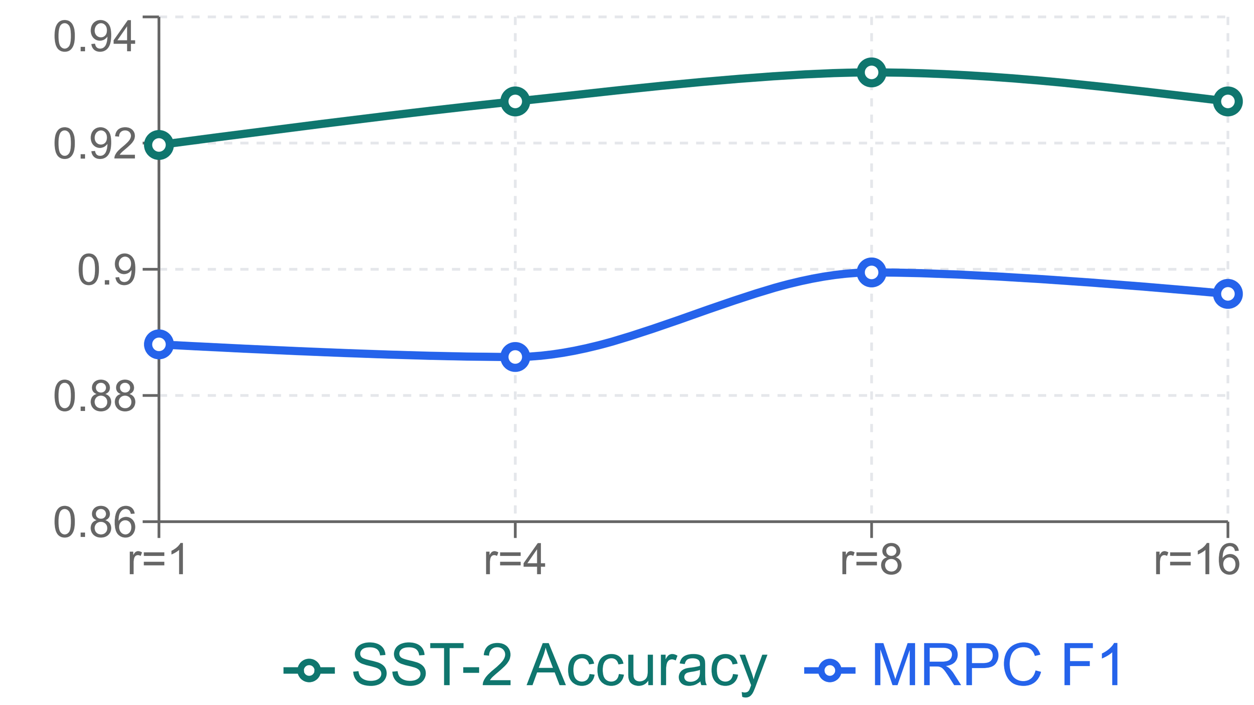
Methodology



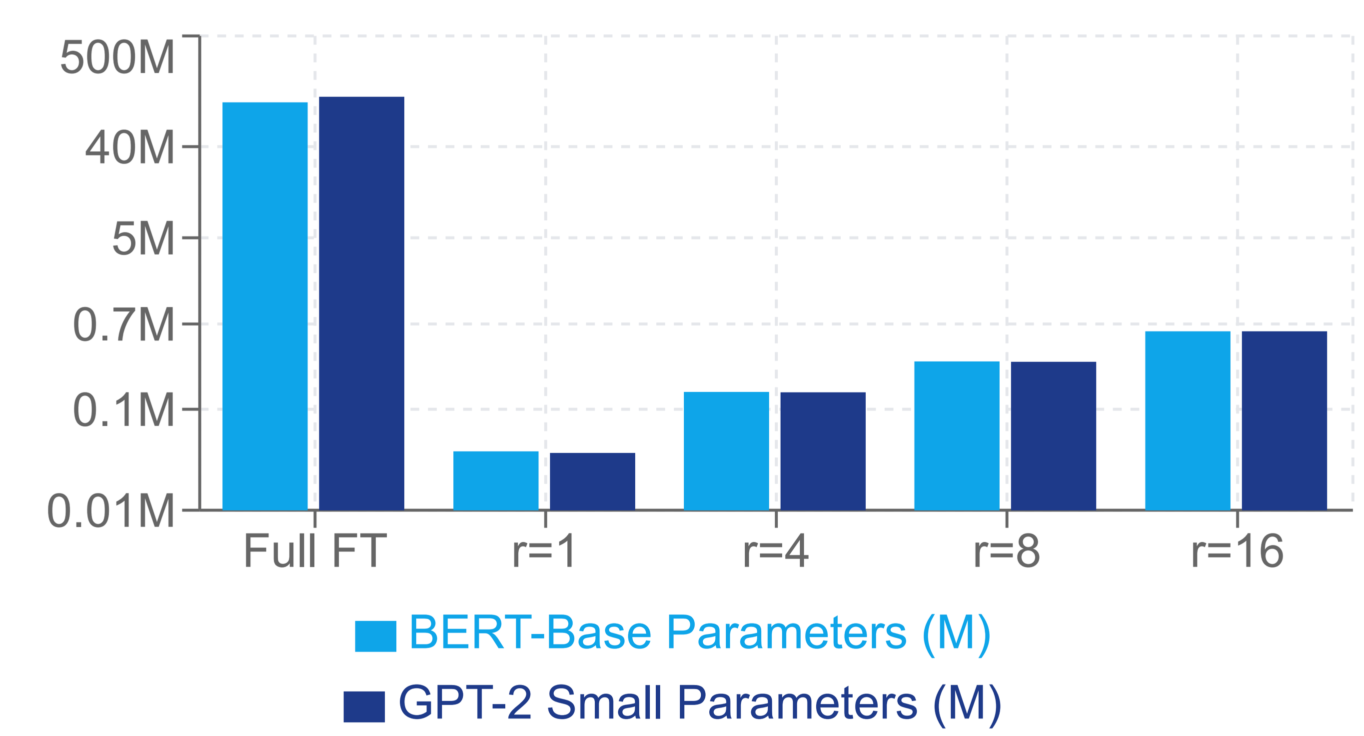
SST-2	"The company reported higher profits this quarter."	Output: Sentiment (Positive)
MRPC	A: "Company profits rose this quarter" B: "The firm reported higher earnings this quarter"	Output: Paraphrased? (Yes)
Wikitext-2	"The capital of France is"	Output: Next token (Paris)

Results

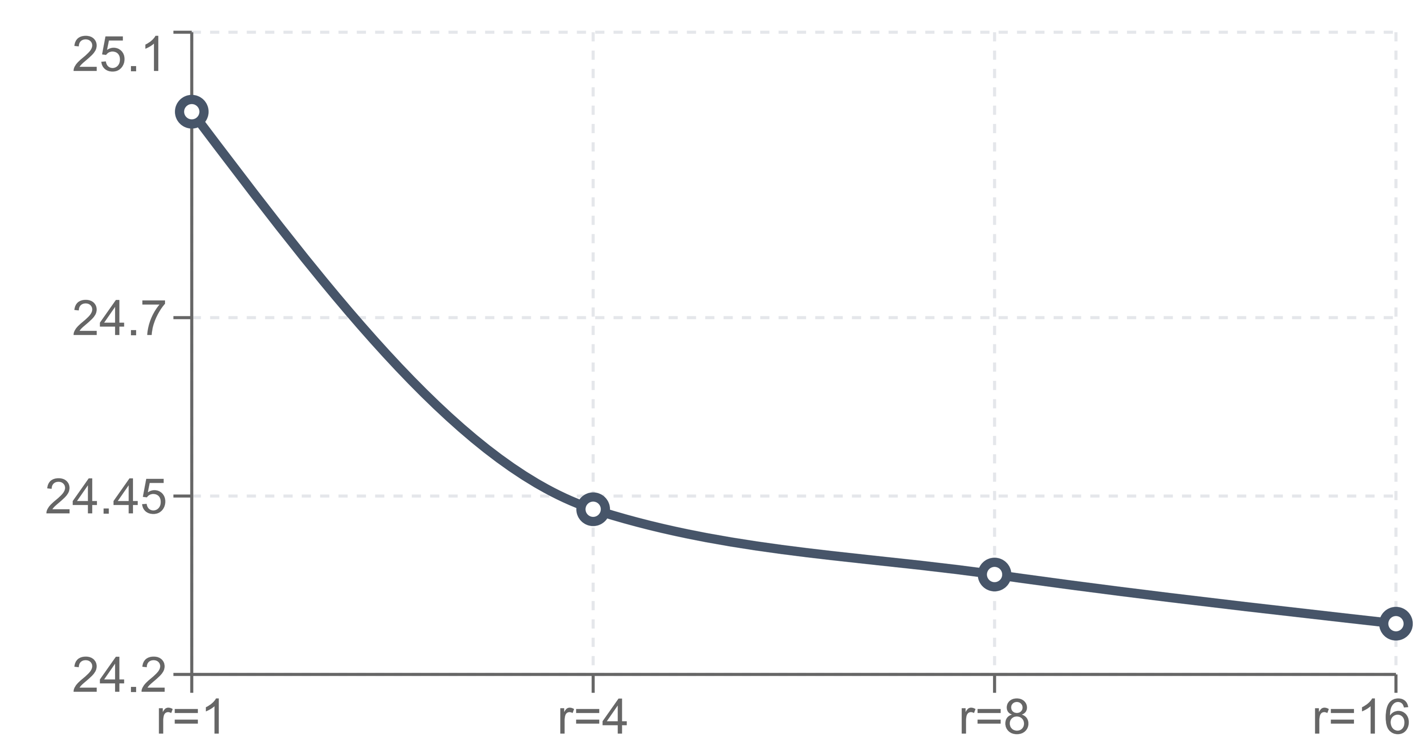
Performance vs LoRA Rank (SST-2, MRPC)



Trainable Parameters vs Rank (Log Scale)



GPT-2 Perplexity vs Rank (WikiText-2)



- **For the future**
  - Can we learn different ranks per layer?
  - Can we adaptively change our rank over time?
  - How does this perform on Vision tasks?

Citations:
Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., & Chen, W. (2022). LoRA: Low-Rank Adaptation of Large Language Models. International Conference on Learning Representations (ICLR). https:/arxiv.org/abs/2106.09685
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